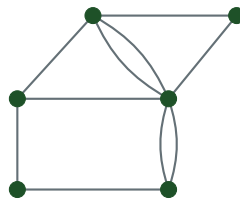


CS-5136: Graph Theory

Operations on Graphs

*pages 51–60 · union, intersection, ring sum &
Cartesian product — with counting proofs and a
direct link back to your reference book's matching theory*



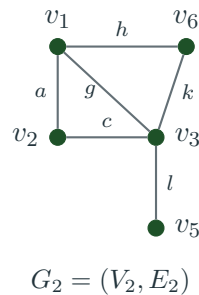
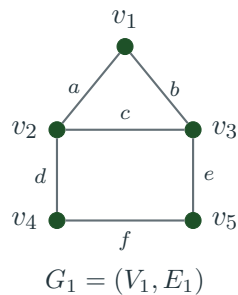
Built from: A. Bhattacharjee, “CS-5136: Graph Theory”, NIT Silchar
with cross-references to S. Held, “Combinatorial Optimization”, Uni Bonn (your reference book)

1 Why Operations on Graphs?

Every kind of mathematical object comes with operations for combining or transforming it into new objects of the same kind: numbers can be added, functions composed, sets intersected. Graphs are no different — given two graphs, we'd like standard ways to combine them into a new graph. Today covers the four such operations from your slides: **union**, **intersection**, **ring sum**, and **Cartesian product**.

Running example

Every operation below is applied to the *same* pair of graphs G_1, G_2 , so you can watch how each one changes. G_1 and G_2 deliberately share some vertices (v_1, v_2, v_3, v_5) and even some *identically-named* edges (a, c) — that overlap is exactly what makes the four operations produce different results from each other.



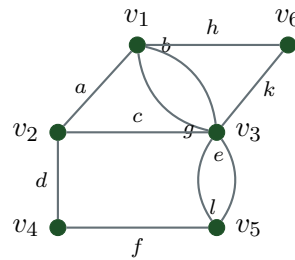
2 The Four Operations

2.1 Union

Definition 2.1 (Union)

The **union** of $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the graph $G_3 = G_1 \cup G_2$ with $V_3 = V_1 \cup V_2$ and $E_3 = E_1 \cup E_2$.

EXAMPLE — $G_3 = G_1 \cup G_2$



$V_3 = \{v_1, \dots, v_6\}$ (6 vertices: the shared v_1, v_2, v_3, v_5 plus G_1 's private v_4 and G_2 's private v_6). E_3 has 10 edges: a and c are the *same* edge in both graphs, so they appear once each; but b (from G_1) and g (from G_2) are *different* edges that happen to join the same pair v_1, v_3 — so in the union they survive as a pair of **parallel edges**. Likewise e and l become parallel edges between v_3, v_5 .

2.2 Intersection

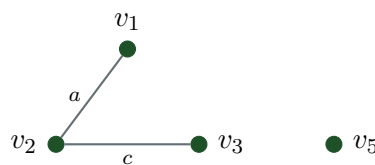
Definition 2.2 (Intersection)

The **intersection** $G_1 \cap G_2$ is the graph G_4 consisting only of the vertices and edges that belong to *both* G_1 and G_2 .

The subtlety worth catching

“Belongs to both” is about edge *identity* (the label), not just endpoints. $e = (v_3, v_5)$ is in G_1 and $l = (v_3, v_5)$ is in G_2 — same endpoints, but they are *different* edges, so *neither* survives into the intersection. Only a and c — literally the same edge in both graphs — do.

EXAMPLE — $G_4 = G_1 \cap G_2$



$V_4 = \{v_1, v_2, v_3, v_5\}$ (every vertex common to both — v_5 is included since $v_5 \in V_1 \cap V_2$, even though it ends up **isolated**, since neither of its edges survived). $E_4 = \{a, c\}$.

2.3 Ring sum

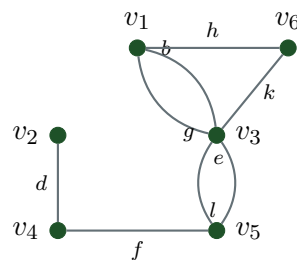
Definition 2.3 (Ring sum)

The **ring sum** $G_1 \oplus G_2$ is the graph G_5 with $V = V_1 \cup V_2$ and edge set equal to those edges in G_1 or G_2 , **but not both**.

You've already met this operation

$E(G_1 \oplus G_2) = (E_1 \setminus E_2) \cup (E_2 \setminus E_1)$ — exactly the **symmetric difference** of the two edge sets. This is precisely the $\triangle$ operation from Held's Lemma 1.6 in your Combinatorial Optimization reference book, where $M \triangle P = (M \setminus P) \cup (P \setminus M)$ for a matching M and a path P ! Same operation, different objects: there it combined a matching with a path; here it combines two whole graphs.

EXAMPLE — $G_5 = G_1 \oplus G_2$



Same vertex set as the union (6 vertices), but a and c — the edges common to both — have vanished, leaving 8 edges: $\{b, d, e, f, g, h, k, l\}$. Vertex v_2 keeps only d ; it lost a .

2.4 Cartesian product

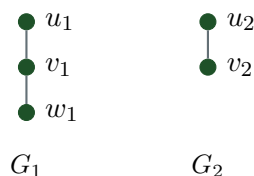
Definition 2.4 (Cartesian product)

The **Cartesian product**^a $G_1 \boxtimes G_2$ is the graph G with $V = V_1 \times V_2$, where distinct vertices (u, v) and (x, y) are adjacent exactly when **either** (1) $u = x$ and $vy \in E(G_2)$, **or** (2) $v = y$ and $ux \in E(G_1)$.

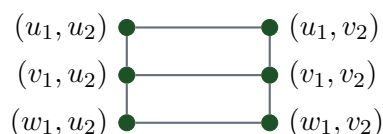
^aMost textbooks write this as $G_1 \square G_2$; your slides use $G_1 \boxtimes G_2$ for the same thing — just be aware $\boxtimes$ means something different (the *strong* product) in most other sources.

EXAMPLE — $P_3 \boxtimes P_2$

Let G_1 be the path $u_1 - v_1 - w_1$ and G_2 the path $u_2 - v_2$.



The product has $3 \times 2 = 6$ vertices, one per pair, forming a 3×2 grid (a “ladder”):



Horizontal edges come from rule (1) (same G_1 -coordinate, an edge in G_2); vertical edges from rule (2) (same G_2 -coordinate, an edge in G_1). Counting: 3 horizontal + 4 vertical = 7 edges — matches the formula proved in Theorem 3.3 below.

3 Provable Properties

None of this is stated as a theorem in your slides, but every one of these is a natural (and examinable) fact about the four operations, and every proof is short. $n_i = |V_i|$, $m_i = |E_i|$ throughout.

Theorem 3.1 (Counting for union and intersection)

$$|V(G_1 \cup G_2)| = n_1 + n_2 - |V_1 \cap V_2|, \quad \|G_1 \cup G_2\| = m_1 + m_2 - \|G_1 \cap G_2\|.$$

Proof. This is just inclusion–exclusion for finite sets: for any two finite sets A, B , $|A \cup B| = |A| + |B| - |A \cap B|$ (elements of $A \cap B$ get counted once in $|A|$ and once in $|B|$, so we subtract the double-count once). Apply this with $A = V_1, B = V_2$ for the first identity, and $A = E_1, B = E_2$ for the second. ■

EXAMPLE

$n_1 = n_2 = 5$, $|V_1 \cap V_2| = |\{v_1, v_2, v_3, v_5\}| = 4 \Rightarrow |V_3| = 5 + 5 - 4 = 6$. ✓ $m_1 = m_2 = 6$, $\|G_1 \cap G_2\| = |\{a, c\}| = 2 \Rightarrow \|G_3\| = 6 + 6 - 2 = 10$. ✓

Theorem 3.2 (Counting for ring sum)

$$\|G_1 \oplus G_2\| = m_1 + m_2 - 2\|G_1 \cap G_2\|.$$

Proof. $E(G_1 \oplus G_2) = E_1 \triangle E_2$, the symmetric difference. For any finite sets, $|A \triangle B| = |A \cup B| - |A \cap B|$ (everything in the union except what's in both), and $|A \cup B| = |A| + |B| - |A \cap B|$ (Theorem 3.1). Substituting:

$$|A \triangle B| = (|A| + |B| - |A \cap B|) - |A \cap B| = |A| + |B| - 2|A \cap B|.$$

Apply with $A = E_1, B = E_2$. ■

EXAMPLE

$\|G_1 \oplus G_2\| = 6 + 6 - 2(2) = 8$, matching $\{b, d, e, f, g, h, k, l\}$ found above. ✓

Theorem 3.3 (Counting for Cartesian product)

$$|V(G_1 \boxtimes G_2)| = n_1 n_2, \quad \|G_1 \boxtimes G_2\| = n_1 m_2 + n_2 m_1.$$

Proof. The vertex count is immediate: $V = V_1 \times V_2$ has exactly $n_1 n_2$ elements.

For edges, split them by which rule of Definition 2.4 created them. **Rule-(1) edges** fix the first coordinate ($u = x$) and vary the second according to $E(G_2)$: for each of the n_1 choices of that fixed first coordinate, we get one copy of all of G_2 's edges, so $n_1 m_2$ edges this way. **Rule-(2) edges** symmetrically fix the second coordinate and vary the first according to $E(G_1)$, giving $n_2 m_1$ edges. These two families never overlap: a rule-(1) edge changes only the second coordinate, a rule-(2) edge changes only the first, and an edge connecting two *distinct* vertices can't do both at once. So the totals simply add: $\|G_1 \boxtimes G_2\| = n_1 m_2 + n_2 m_1$. ■

EXAMPLE

$n_1 = 3, m_1 = 2, n_2 = 2, m_2 = 1$: vertices $= 3 \times 2 = 6$ ✓; edges $= n_1 m_2 + n_2 m_1 = 3(1) + 2(2) = 3 + 4 = 7$ ✓ — matches the ladder graph drawn above exactly.

Two quick remarks

Union, intersection, and ring sum are all built directly from *set* union/intersection/symmetric-difference on V_i and E_i — so they instantly inherit commutativity ($G_1 \cup G_2 = G_2 \cup G_1$, etc.) from the corresponding set identities. The Cartesian product is commutative too, but only up to *isomorphism*: swapping coordinates, $(u, v) \mapsto (v, u)$, is a bijection $V_1 \times V_2 \rightarrow V_2 \times V_1$ that preserves adjacency, so $G_1 \boxtimes G_2 \cong G_2 \boxtimes G_1$.

4 Day 3 Cheat-Sheet

Recap

- Union: $V_1 \cup V_2, E_1 \cup E_2$. Same-endpoint edges with *different* names survive as parallel edges.
- Intersection: only vertices/edges in *both* — edge identity matters, not just endpoints.
- Ring sum: symmetric difference of edge sets — literally the same $\triangle$ as Held's $M \triangle P$.
- Cartesian product: $V_1 \times V_2$; adjacency = “move along one coordinate the way the corresponding graph allows.”
- Counting: $|V_1 \cup V_2| = n_1 + n_2 - |V_1 \cap V_2|$; $\|G_1 \oplus G_2\| = m_1 + m_2 - 2\|G_1 \cap G_2\|$; $\|G_1 \boxtimes G_2\| = n_1 m_2 + n_2 m_1$.

5 Practice Questions

Set A --- straight from today's definitions

Q1. Using G_1, G_2 from Section 1: is d (the edge (v_2, v_4)) present in $G_1 \cap G_2$? Why or why not?

A. No. $v_4 \notin V_2$ at all, so d can't possibly be an edge of G_2 , let alone a shared one.

Q2. Compute $d(v_3)$ (the degree of v_3) in G_1 , in G_2 , and in $G_1 \cup G_2$. Is it just the sum of the first two?

A. In G_1 : $d(v_3) = 3$ (b, c, e). In G_2 : $d(v_3) = 4$ (g, k, c, l). Naively summing gives 7, but the union only has degree 6 ($\{b, c, e, g, k, l\}$) — one less, because c is the same edge in both graphs and got counted twice in $3 + 4$. Moral: degrees add directly only when no edge at that vertex is shared between the two graphs; otherwise subtract one for each shared edge at that vertex.

Q3. Draw $P_2 \boxtimes P_2$ (two copies of a single edge). How many vertices and edges does it have, and what shape is it?

A. $n_1 = n_2 = 2, m_1 = m_2 = 1$. Vertices: $2 \times 2 = 4$. Edges: $n_1 m_2 + n_2 m_1 = 2(1) + 2(1) = 4$. Four vertices, four edges, each vertex degree 2 — that's the cycle graph C_4 (a square).

Set B --- properties

Q4. If G_1 and G_2 have completely disjoint vertex sets (no shared vertices at all, so certainly no shared edges), what do $G_1 \cap G_2$, $G_1 \cup G_2$, and $G_1 \oplus G_2$ look like?

A. $G_1 \cap G_2$ is the null graph (no vertices, no edges) — or depending on convention, simply undefined/empty. $G_1 \cup G_2$ and $G_1 \oplus G_2$ coincide exactly (since $E_1 \cap E_2 = \emptyset$ means nothing gets removed): both equal the graph with $V_1 \cup V_2$ and $E_1 \cup E_2$, i.e. G_1 and G_2 drawn side by side with no interaction.

Q5. Using Theorem 3.1: if $\|G_1 \cup G_2\| = 9, m_1 = 6, m_2 = 5$, what is $\|G_1 \cap G_2\|$?

A. $9 = 6 + 5 - \|G_1 \cap G_2\| \Rightarrow \|G_1 \cap G_2\| = 11 - 9 = 2$.

Set C --- bridging to your reference book (Held, *Combinatorial Optimization*)

Q6. Held's Lemma 1.6 computes $M \triangle P$ for a matching M and an M -alternating path P , both viewed as *sets of edges* of the same graph G . Using today's Theorem 3.2's formula with $A = M, B = E(P)$: if $|M| = 4$, P has 5 edges, and M, P share 2 edges in common, how many edges does $M \triangle P$ have? Does this match what Lemma 1.6 predicts for an augmenting path (recall: $|M \triangle P| = |M| + 1$ when P is M -augmenting)?

A. $|M \triangle P| = |M| + |E(P)| - 2|M \cap E(P)| = 4 + 5 - 2(2) = 5$. Held's Lemma 1.6 says an M -augmenting

path of length ℓ (i.e. ℓ edges) has exactly $\lceil \ell/2 \rceil$ non- M edges and $\lfloor \ell/2 \rfloor$ M -edges, with $|M \triangle P| = |M| + 1$. Here P has 5 edges with 2 shared with M (so 3 non- M edges, 2 M -edges) — exactly the alternating pattern of a length-5 augmenting path, and indeed $5 = 4 + 1 = |M| + 1$. ✓ Same formula, same conclusion, two different books.

Q7. The proof of Held’s Theorem 1.7 (Berge’s Augmenting Path Theorem, Day 1 of the Combinatorial Optimization deck) builds $H = M \triangle M'$ from two matchings M, M' and studies its structure. Using G_2 from Day 2 (5 vertices, edges $(v_1, v_2), (v_1, v_3), (v_2, v_4), (v_3, v_4), (v_4, v_5)$): let $M = \{(v_1, v_2), (v_3, v_4)\}$ and $M' = \{(v_1, v_3), (v_2, v_4)\}$. Compute $M \cup M'$, $M \cap M'$, and $M \triangle M'$ as edge sets, then check Theorem 3.2’s count against direct listing.

A. $M \cap M' = \emptyset$ (no shared edges). $M \cup M' = \{(v_1, v_2), (v_3, v_4), (v_1, v_3), (v_2, v_4)\}$, size 4. $M \triangle M'$: since the intersection is empty, $M \triangle M' = M \cup M'$ too — all 4 edges. Theorem 3.2: $|M| + |M'| - 2|M \cap M'| = 2 + 2 - 2(0) = 4$. ✓ (This $H = M \triangle M'$ is in fact a single 4-cycle $v_1 v_2 v_4 v_3 v_1$ — exactly the “even alternating cycle” shape that Theorem 1.7’s proof decomposes H into, Day 1 of the Combinatorial Optimization deck.)

Just say “Day 4” whenever you’re ready for the next set (pages 61 onward, Algebraic treatment: isomorphism and beyond).